

Math 150A Section 8 and 16
Spring 2025
Exam III
April 17, 2025
Time Limit: 1 Hour 50 Minutes

Name (Print): _____

Student ID: _____

This exam contains 8 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam. However, you may use a single, handwritten, one-sided notesheet and a *basic* calculator.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total:	70	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** Write

(a)

$$\int \cos(2x)dx = \sin(2x) + C$$

(b)

$$\sum_{k=1}^3 k \cos(\pi k) = 6$$

(c)

$$1^2 + 2^2 + 3^2 + 4^2 + \cdots + (99)^2 = \frac{99 \cdot 100 \cdot 199}{6}$$

(d) if $\int_0^1 f(x)dx = 1$ and $\int_0^2 f(x)dx = 3$ then $\int_1^2 3f(x)dx = 6$

(e) assuming f and g are continuous

$$\int f(x)g(x)dx = \left(\int f(x)dx\right)g(x) + f(x)\left(\int g(x)dx\right)$$

2. (10 points)

(a) Calculate $h'(x)$ for $h(x)$ the function

$$h(x) = \int_{-x}^{x^2} \frac{t}{t^2 + 1} dt$$

(b) Calculate $\int_{-5}^0 3\sqrt{25 - x^2} \, dx$. Carefully justify your work.

3. (10 points) Use Newton's method to approximate a solution of the equation accurate to four decimal places

$$x^2 - 3 = 0.$$

Explain how you know when to stop and **SHOW YOUR WORK!**

4. (10 points) Compute each of the following integrals. Show all work. Lack of work results in no credit!!

(a) $\int_1^2 4u^3 + 2u + 5 \, du$

(b) $\int (\sqrt{x} - 1) \left(x^2 + \frac{1}{x^2} \right) dx$

(c) $\int_0^{\pi/4} \frac{5z + 3z \cos(z)}{z} \, dz$

(d) $\int \sec^2(t) + \frac{1}{\sqrt[3]{t}} \, dt$

5. (10 points)

A chocolate company is creating a box with a square base. The company only wants to use 100 square centimeters of material per box. What is the largest volume they can make for such a box?

6. (10 points)

(a) Determine an integral whose value is equal to $\lim_{n \rightarrow \infty} R_n$ where here

$$R_n = \sum_{k=1}^n \left(2 + \frac{5k}{n}\right)^3 \frac{5}{n}.$$

(b) Compute the value of the left Riemann sum L_3 for $f(x) = x^5 + 1$ on the interval $[-2, 4]$.

7. (10 points)

Consider the Riemann integral

$$\int_{-1}^2 x^2 + 2x + 1 \, dx$$

(a) Determine an explicit expression for the right Riemann sum R_n

(b) Compute the Riemann integral by calculating the limit $\lim_{n \rightarrow \infty} R_n$. Carefully justify!